

RedHatAI/Qwen3.8-27B-INT4

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  Transformers
  Safetensors
 qwen3_5
 vllm
 llm-compressor
 compressed-tensors
 int4
 conversational
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 Deploy
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 Use this model

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RedHatAI/Qwen3.8-27B-INT4

This model is a quantized version of [Qwen/Qwen3.8-27B](#).

Model Optimizations

This model was obtained by quantizing the weights of [Qwen/Qwen3.8-27B](#) to int4, ready for inference with vLLM. Weights are quantized to int4 with a group size of 128. Only the weights of the linear operators within transformer blocks are quantized using [LLM Compressor](#). Vision tower and output head layers are kept in their original precision.

Creation Code

```
import torch
from compressed_tensors.quantization.quant_s
from datasets import load_dataset
from transformers import AutoProcessor, Qwen

from llmcompressor import oneshot
from llmcompressor.modifiers.gptq import GPTQ
from llmcompressor.modifiers.transform.awq import
from llmcompressor.utils import load_context
```

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105,186


Safetensors

Model size 27B params

Tensor type I32 · BF16

 Chat template
  Files info

Inference Providers **NEW**

 Image-Text-to-Text

This model isn't deployed by any Inference Provider.

 Ask for provider support

Model tree for RedHatAI/Qwen3.8-...

 Base model [Qwen/Qwen3.8-27B](#)
 Quantized (821) [this model](#)

```
MODEL_ID = "Qwen/Qwen3.8-27B"
```

```
# Load model.
```

```
with load_context(Qwen3_5ForConditionalGenera  
    model = Qwen3_5ForConditionalGeneration.:  
processor = AutoProcessor.from_pretrained(MOI
```

```
recipe = [  
    AWQModifier(duo_scaling="both"),  
    GPTQModifier(  
        targets="Linear",  
        scheme="W4A16",  
        ignore=[  
            "re:visual.*",  
            "re:model.visual.*",  
            r"re:.*lm_head",  
            "re:.*embed_tokens$",  
            r"re:.*linear_attn\.in_proj_a$",  
            r"re:.*linear_attn\.in_proj_b$",  
        ],  
    )  
]
```

```
NUM_CALIBRATION_SAMPLES = 512
```

```
MAX_SEQUENCE_LENGTH = 4096
```

```
ds = load_dataset(  
    "mlabonne/open-perfectblend",  
    split=f"train[:{NUM_CALIBRATION_SAMPLES}]"  
)  
ds = ds.shuffle(seed=42)
```

```
ROLE_MAP = {"human": "user", "gpt": "assistant"}
```

```
def preprocess_function(example):  
    messages = [  
        {  
            "role": ROLE_MAP.get(msg["from"]
```

```

        "content": [{ "type": "text", "text": msg["text"] }
    ]
    for msg in example["conversations"]
]
return processor.apply_chat_template(
    messages,
    tokenize=True,
    return_dict=True,
    add_generation_prompt=False,
    processor_kwargs={
        "return_tensors": "pt",
        "padding": False,
        "truncation": True,
        "max_length": MAX_SEQUENCE_LENGTH,
        "add_special_tokens": False,
    },
)

```

```
ds = ds.map(preprocess_function, batched=False)
```

```

def data_collator(batch):
    assert len(batch) == 1
    return {key: torch.tensor(value) for key, value in batch[0].items()}

```

Apply quantization.

```

oneshot(
    model=model,
    recipe=recipe,
    dataset=ds,
    max_seq_length=MAX_SEQUENCE_LENGTH,
    num_calibration_samples=NUM_CALIBRATION_SAMPLES,
    moe_calibrate_all_experts=True,
    data_collator=data_collator,
)

```

Save to disk in compressed-tensors format.

```
SAVE_DIR = MODEL_ID.rstrip("/").split("/")[-1]
```

```
model.save_pretrained(SAVE_DIR)
processor.save_pretrained(SAVE_DIR)
```

vLLM Serving

```
vllm serve RedHatAI/Qwen3.8-27B-INT4
--tensor-parallel-size 1
--enable-auto-tool-choice
--tool-call-parser qwen3_coder
--reasoning-parser qwen3
--mm-encoder-tp-mode data
--max-model-len 69632
--gpu-memory-utilization 0.9
```

Evaluations

Eval	Qwen/Qwen3.8-27B (BF16)	RedHatAI/Qwen3.8-27B (BF16)
gsm8k_platinum_cot_llama	0.9589	0.9677
ifeval	0.9187	0.9211
aime25	0.9375	0.9542
math_500	0.8360	0.8380
gpqa_diamond	0.8923	0.8636

Eval commands

Run once per seed (1234, 2345, 3456). Final reported values are averages.

gsm8k_platinum_cot_llama

```
lm_eval \
  --model local-chat-completions \
  --tasks gsm8k_platinum_cot_llama \
  --model_args model=Qwen/Qwen3.8-27B,max_length=1024,tokenizer_backend=None,tokenization_kwargs={'padding_side': 'left', 'truncation_side': 'left'},_requests=False,tokenizer_backend=None,timeouts={'completion': 300},\
  --num_fewshot 0 \
  --seed <SEED> \
  --gen_kwargs do_sample=True,temperature=1.0 \
  --apply_chat_template
```

ifeval

```
lm_eval \
  --model local-chat-completions \
  --tasks ifeval \
  --model_args model=Qwen/Qwen3.8-27B,max_length=1024,tokenizer_backend=None,tokenization_kwargs={'padding_side': 'left', 'truncation_side': 'left'},_requests=False,tokenizer_backend=None,timeouts={'completion': 300},\
  --num_fewshot 0 \
  --seed <SEED> \
  --gen_kwargs do_sample=True,temperature=1.0 \
  --apply_chat_template
```

aime25

```
lighteval endpoint litellm \
  runs/20260818_110422_qwen3.8-27b-int4_full. \
  'aime25|0' \
  --save-details
```

math_500

```
lighteval endpoint litellm \
  runs/20260818_110422_qwen3.8-27b-int4_full. \
  'math_500|0' \
  --save-details
```

```
lighteval endpoint litellm \
  runs/20260818_110422_qwen3.8-27b-int4_full.
  'gpqa:diamond|0' \
  --save-details
```